MULTIMODAL AI · CHEST DISEASE SCREENING

AI-Assisted Multimodal Screening of Chest Diseases

Combining clinical information with chest X-ray images



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Problem Statement

"Preliminary disease assessment relies heavily on individual clinical evaluation, while symptoms and medical images are not always analysed together."

Need: an AI-assisted multimodal system that fuses clinical information with chest X-rays to screen selected chest diseases.



Depends on the individual

Screening quality varies with clinician experience and workload.



Data analysed in silos

Symptoms/history and X-ray findings are rarely interpreted together.



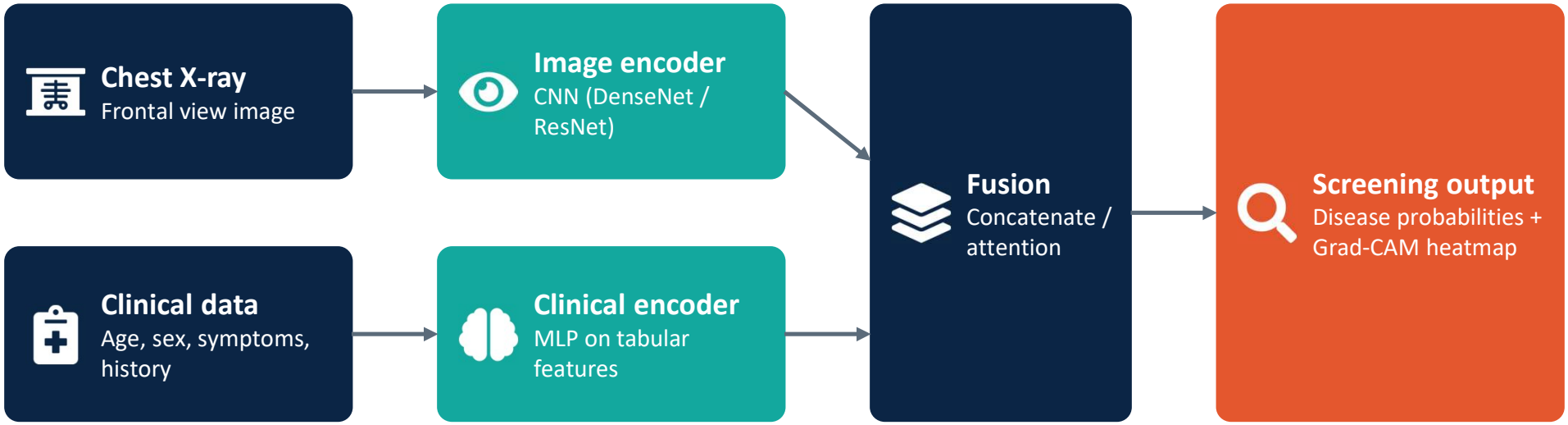
Delays in early screening

Manual reading of every image is slow, especially where radiologists are scarce.

Existing Approaches & Their Limitations

EXISTING APPROACH		LIMITATION
Manual radiologist reading of chest X-rays	→	Time-consuming; inter-observer variability; radiologist shortage
Clinical evaluation of symptoms alone	→	Subjective; cough, fever, breathlessness overlap across many diseases
Image-only deep learning (e.g., CheXNet, DenseNet CNNs)	→	Ignores patient context; more false positives; “black-box” output
Symptoms and images reviewed separately	→	No joint reasoning; fragmented information; slower decisions

Proposed System: Multimodal Fusion



Richer context
Symptoms narrow down look-alike X-ray patterns

Explainable
Heatmaps show where the model looked

Screening aid
Supports, not replaces, the clinician

Methodology Workflow

1

Collect & preprocess

Resize, normalise, augment X-rays; encode and impute clinical features

2

Extract features

CNN backbone for images; MLP for clinical variables

3

Fuse modalities

Concatenate or attention-weight both feature vectors

4

Train & validate

Multi-label loss; patient-wise train/val/test split

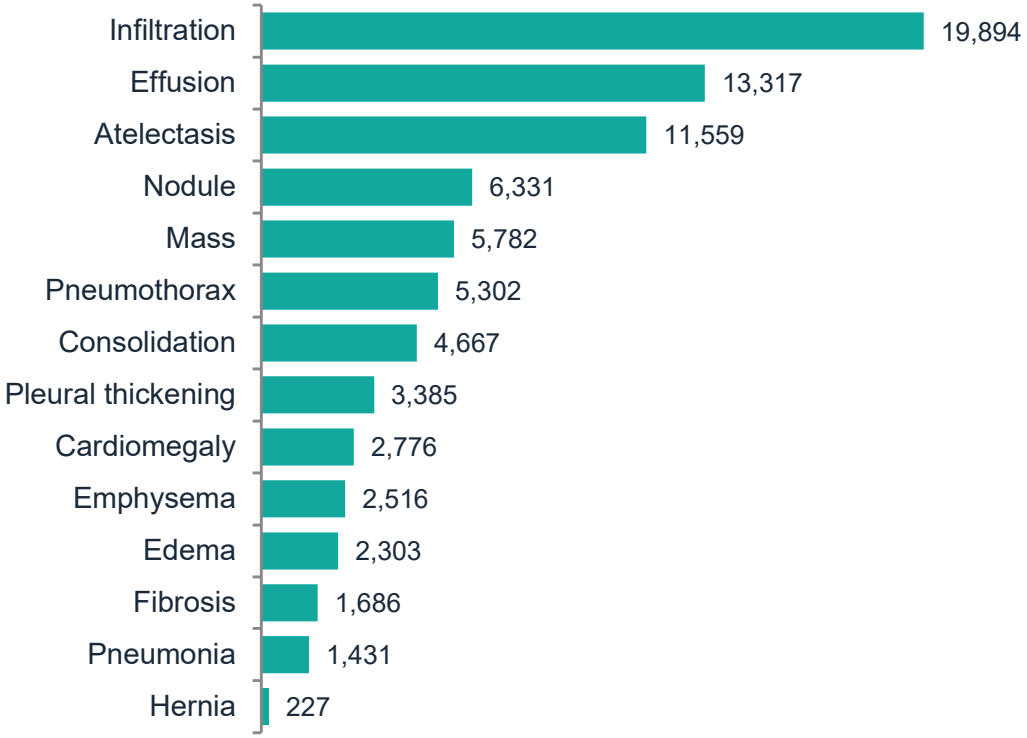
5

Explain & output

Probabilities per disease with Grad-CAM localisation

Dataset Used

NIH ChestX-ray14 — image counts per label (an image can carry several labels)



112,120 frontal-view chest X-rays

30,805 unique patients

14 thoracic disease labels

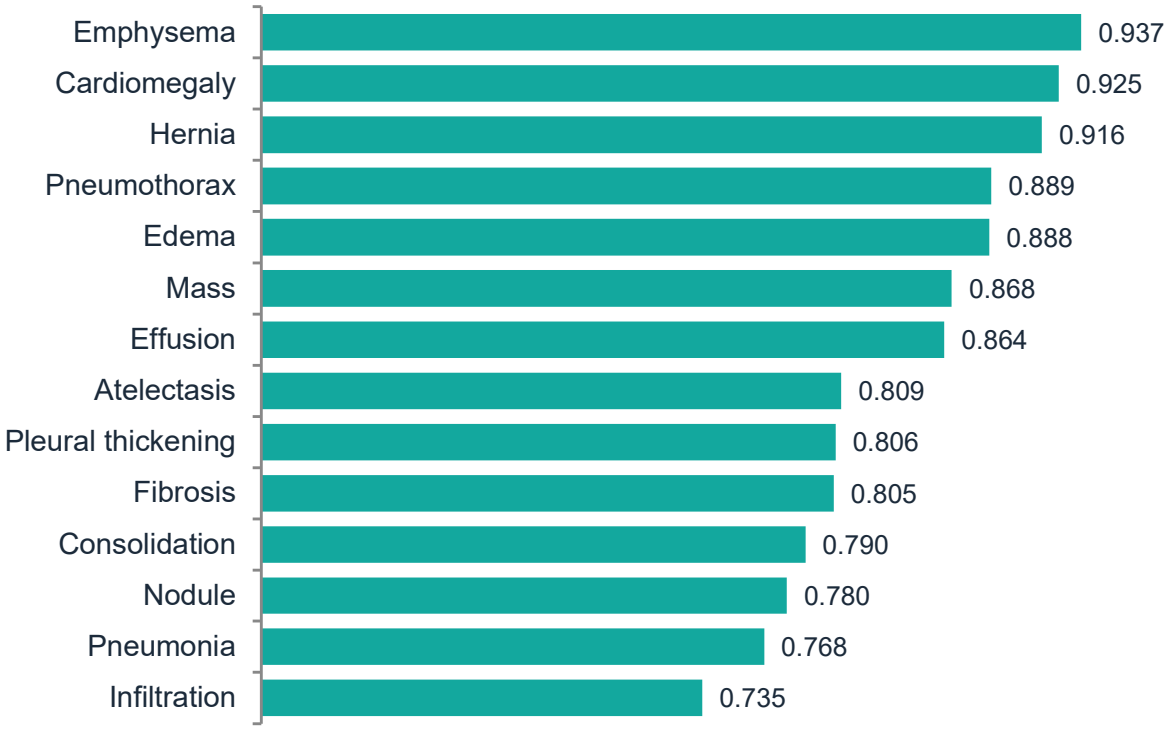
Labels are text-mined from radiology reports.
Clinical inputs: age, sex, view position from metadata, plus symptoms/history.

Research Papers Studied

Paper	Key contribution
Wang et al., CVPR 2017	ChestX-ray8/14: large-scale CXR dataset and multi-label benchmarks
Rajpurkar et al., 2017	CheXNet: 121-layer DenseNet reaching radiologist-level pneumonia detection
Irvin et al., AAAI 2019	CheXpert: 224K CXRs with uncertainty labels and expert comparison
Johnson et al., Sci. Data 2019	MIMIC-CXR: CXRs linked with free-text reports and clinical records
Huang et al., CVPR 2017	DenseNet: dense connectivity — the common image backbone
Selvaraju et al., ICCV 2017	Grad-CAM: visual explanations from CNN predictions
Huang et al., npj Digit. Med. 2020	Review of fusing imaging with EHR data; early vs. late fusion
Hayat et al., ML4H 2022	MedFuse: fusing clinical time-series with chest X-rays

Literature Insight: CheXNet Results

Per-pathology AUROC on ChestX-ray14 test set (re-plotted from Rajpurkar et al., 2017)



0.841

mean AUROC across all 14 pathologies

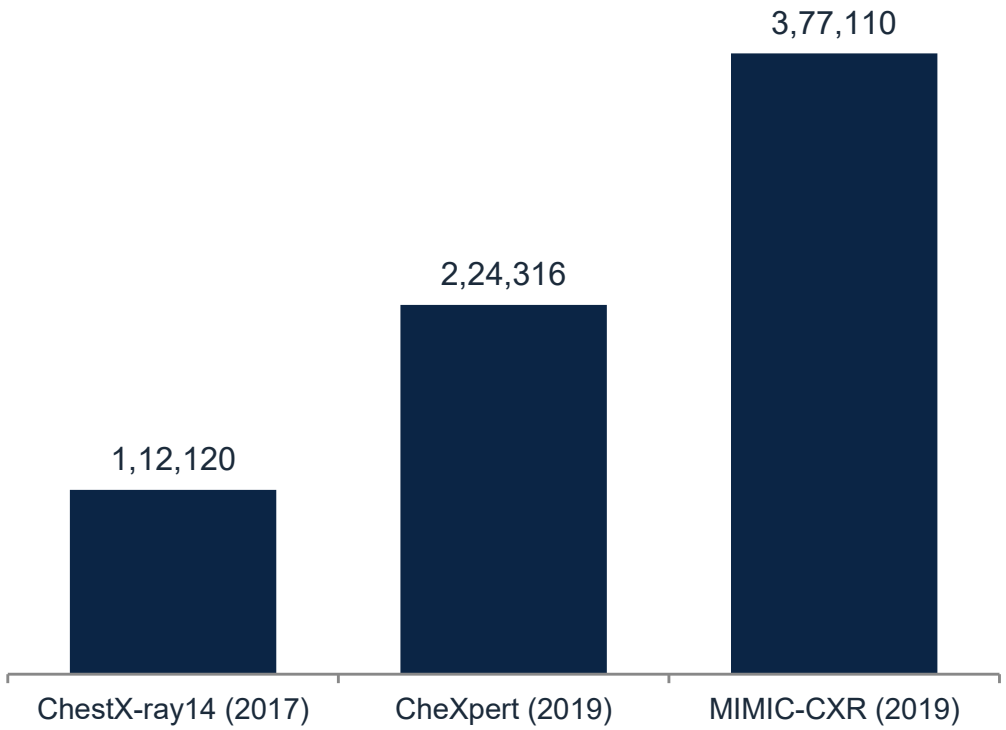
0.435 vs 0.387

pneumonia F1: CheXNet vs. average radiologist

Takeaway: image-only models do well on some findings (emphysema, cardiomegaly) but struggle with infiltration and pneumonia — where clinical context can help.

Literature Insight: Data Scale & Gaps

Number of chest X-ray images in major public datasets



Gaps identified

Most benchmarks are image-only
Clinical variables are seldom modelled with the X-ray.

Label noise & uncertainty
Report-mined labels are imperfect; uncertain findings need handling.

Limited explainability
Clinicians need to see why a prediction was made.

Evaluation Plan & Expected Outcomes

AUROC	Sensitivity	Specificity	F1-score
Per-disease discrimination	Catch true disease cases	Limit false alarms	Precision–recall balance

Ablation design

- Image-only baseline (CNN)
- Clinical-only baseline (MLP)
- Proposed multimodal fusion model
- Compare with paired significance tests

Expected outcomes

- Fusion outperforms either modality alone
- Fewer false positives on look-alike findings
- Heatmaps that align with radiologist attention

Applications, Challenges & Ethics



Applications

- Primary-care and rural triage
- Second-reader support for radiologists
- Prioritising urgent cases in worklists
- Large-scale screening programmes



Challenges & Ethics

- Dataset bias and label noise
- Generalisation across hospitals/devices
- Patient privacy and data consent
- A screening aid, not a diagnosis

Conclusion & Future Scope

- Combining symptoms with chest X-rays gives a more complete screening signal than either alone.
- Fusion + Grad-CAM offers accuracy and explainability for selected chest diseases.
- Future: clinical notes via NLP, multi-hospital validation, lightweight deployment for low-resource clinics.



Thank you · Questions?